

Yajat Mehta

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MASTER PROFILE

Business Data Analytics and Artificial Intelligence in Business student with experience across data analytics, AI applications, cloud-based machine learning, product development, business strategy, systems analysis, and IT governance. This master resume is a full project inventory designed to support targeted resumes for data analyst, AI/ML, product, strategy, cybersecurity/GRC, and emerging technology roles.

EDUCATION

Arizona State University

B.S., Business Data Analytics | B.S., Artificial Intelligence in Business | GPA: 3.56

August 2022 – May 2026

Tempe, AZ

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, HTML, CSS, Swift

Data & ML: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, NLP, RAG, Recommender Systems, Text Classification, Word2Vec, Random Forest, Logistic Regression

Analytics & BI: Tableau, Power BI, Advanced Excel, PivotTables, Dashboarding, Data Visualization, Data Mining, Data Governance

Cloud & Big Data: Google Cloud Platform, Google Cloud Storage, PySpark, Spark ML, AWS ML/NLP Foundations

Backend & Databases: FastAPI, Flask, Supabase, PostgreSQL, Redis, REST APIs, Google OAuth, Docker, Postman

Systems & Modeling: SDLC, Requirements Analysis, Use Cases, DFDs, ERDs, Schema Design, Dimensional Modeling, Star Schema, Snowflake Schema, Data Warehousing

Security & Governance: FERPA-aware documentation, Access Control, Risk Mitigation, BCP/DRP, COBIT, ISO, NIST, Compliance Planning

Tools: Git, GitHub, VS Code, Figma, Xcode, Google Colab

Certifications: The Global Career Accelerator: Tableau, Advanced Excel; AWS: Machine Learning for Natural Language Processing; AWS Academy: Machine Learning Foundations

PROFESSIONAL EXPERIENCE

Founder & Developer, Rumore — Campus Social Platform

2025 – Present

FastAPI, Supabase, PostgreSQL, Redis, React Native/Expo, Google OAuth

Remote / Tempe, AZ

- Built a mobile-first social platform for college communities featuring anonymous campus posts, identity-separated user flows, and scoped authentication for privacy-sensitive student interaction.
- Developed backend architecture using FastAPI, Supabase PostgreSQL, Redis, and custom Google OAuth flows to support secure login, token handling, session control, and campus-based access control.
- Designed platform architecture around privacy-sensitive student communities, moderation workflows, protected API access, and future campus expansion.
- Prepared product strategy for launch in Indian colleges after delaying U.S. publication, adapting the roadmap toward campus-specific communities and scalable engagement.

Data Analyst Intern, Mehta Papers Pvt. Ltd.

May 2023 – August 2023

Excel, Tableau, Sales Analysis, Inventory Analysis

Delhi, India

- Analyzed historical and current sales, inventory, pricing, and supplier data using Excel and Tableau to identify seasonal demand trends in wholesale paper distribution.
- Performed demand forecasting and price-trend analysis around school-year purchasing cycles, helping the business anticipate peak inventory needs before high-demand months.
- Cleaned, merged, and structured data from multiple internal and supplier sources to improve data consistency and support more accurate business reporting.
- Created Excel reports and Tableau dashboards tracking sales trends, inventory movement, supplier activity, and pricing changes for management decision-making.

Desk Aide, Arizona State University

August 2023 – February 2026

University Housing Operations, Student Support, Documentation

Tempe, AZ

- Managed residence hall front-desk operations by handling phone calls, walk-in inquiries, visitor support, and student housing service requests.
- Processed key-related services including lockouts, key checkouts, lock-change requests, and new key order documentation.
- Submitted and tracked maintenance requests, housing records, and administrative documentation while following FERPA-related confidentiality standards.
- Coordinated with desk supervisors and Resident Assistants to resolve resident concerns, escalate urgent issues, and support daily housing operations.

AI, MACHINE LEARNING & CLOUD PROJECTS

Marketing Assistant AI Agent

May 2026

- Built a Python-based AI marketing assistant bot that allowed users to query campaign data and receive analysis across performance metrics, audience trends, and business documents.
- Designed the system to connect with existing marketing data sources and consolidate scattered campaign information into a single queryable assistant for faster decision-making.
- Developed an agent workflow for data aggregation, validation, insight generation, report building, and feedback refinement to keep marketers in control of final decisions.
- Evaluated implementation risks including hallucinated metrics, missing data retrieval, data privacy, tool dependency, and accountability in AI-supported marketing workflows.

Sentiment Analysis Using GCP and PySpark

May 2025

- Built a **PySpark** NLP pipeline on **Google Cloud** to classify Amazon review sentiment using tokenization, stopword removal, **Word2Vec** vectorization, and **Random Forest** classification.
- Loaded data from **Google Cloud Storage**, processed text data in Colab/GCP, and saved the trained Spark ML model back to a GCP bucket for reuse.
- Preprocessed text using tokenization, stopword removal, punctuation handling, and train-test splitting to evaluate the impact of text cleaning on model performance.
- Evaluated model performance using F1 score and compared pipeline behavior with and without punctuation removal.

Factors Influencing Heart Disease

December 2024

- Analyzed a Kaggle heart disease dataset using **Python** to identify clinical and demographic factors influencing heart disease risk.
- Cleaned and preprocessed patient data with **Pandas** by removing duplicates, checking missing values, and preparing features for machine learning models.
- Trained and evaluated **Logistic Regression** and **Random Forest** classifiers using an 80/20 train-test split, feature scaling, GridSearchCV, and classification metrics.
- Achieved 82% Random Forest accuracy, 90% recall, and 0.86 AUC, identifying chest pain type, number of major vessels, and maximum heart rate as key predictors.

DATA ANALYTICS, BI & DATA WAREHOUSING PROJECTS

Medicaid Pharmaceutical Spending Analysis

May 2026

- Analyzed Medicaid drug reimbursement and 2024 population datasets to identify high-cost drugs, reimbursement pressure, and state-level spending variation.
- Built **Tableau** dashboards with calculated per-capita reimbursement fields to compare drug costs, Medicaid vs. non-Medicaid spending, and geographic cost concentration.
- Found that high-cost specialty drugs, rather than high-volume generic medications, created the strongest financial pressure, supporting targeted rebate negotiation and prior authorization strategies.
- Identified high-spending states and regional patterns to support state-level cost-control strategies and policy benchmarking.

Snowflake Schema Design for Ticket Sales Data Warehouse

December 2025

- Converted a transaction-grained star schema into a snowflake schema by normalizing dimensional hierarchies and maintaining correct fact-to-dimension relationships.
- Designed SQL CREATE TABLE statements with surrogate keys, primary keys, foreign keys, and hierarchical dimension relationships for ticket sales analysis.
- Modeled a central ticket sales fact table connected to date, attendee, event, venue, city, month, and year dimensions to support structured analytical querying.

Investigating Potential Lending Bias in Mortgage Applications Using HMDA Data

April 2024

- Analyzed 12,550 loan applications to identify patterns of potential lending bias based on race and ethnicity in mortgage approval rates.
- Created PivotTables and Excel dashboards, uncovering that 82.16% of loans were approved while 17.84% were denied.
- Conducted a statistical A/B test showing that middle-class White applicants had a 22% higher approval rate than Black applicants, providing insight into potential systemic lending bias.

Data Center Location Selection Using Energy Surplus Data

April 2024

- Designed interactive **Tableau** dashboards to analyze 11 regions' energy surplus and renewable energy share for data center location decision-making.
- Developed a dual-axis supply and demand chart, identifying August as the peak energy surplus month for the Mid-Atlantic region.
- Identified 3:00 PM as the top wind energy production hour in California, supporting renewable energy planning for potential data center siting.

PRODUCT, SOFTWARE & SYSTEMS PROJECTS

Rumore — Campus Social Platform

2025 – Present

- Built a mobile-first campus social platform with anonymous posting, campus-based access control, and privacy-focused identity separation for student communities.
- Developed backend services using **FastAPI**, **Supabase PostgreSQL**, **Redis**, and **Google OAuth** to support authentication, token handling, session control, and protected API access.
- Designed platform roadmap around campus-specific communities, moderation workflows, and scalable student engagement.

Student Event Management System (SEMS)

2025

- Designed a Student Event Management System to centralize campus event discovery, approvals, RSVP tracking, and attendance logging for students, organizers, and administrators.
- Developed **SDLC** documentation across planning, analysis, design, and implementation phases, including stakeholder analysis, functional requirements, non-functional requirements, use cases, **DFDs**, **ERD**, and UI wireframes.
- Defined role-based workflows for students, organizers, and admins covering event creation, approval, RSVP enforcement, check-in tracking, reporting, and access control.

SwiftUI Map-Based Location App

2023

- Developed an iOS map application in **SwiftUI** using **MapKit** to render a custom user-selected location based on latitude and longitude coordinates.
- Customized map annotations, location labels, and simulator behavior to demonstrate location-based UI control and foundational mobile app development skills.

BUSINESS STRATEGY & CONSULTING PROJECTS

Airbnb Strategic Analysis & Recommendation

April 2026

- Conducted a strategic analysis of Airbnb's trust, safety, fraud, and quality-control challenges, identifying brand fragility as a core risk to long-term competitive advantage.
- Applied **PESTEL**, **Porter's Five Forces**, **VRIO**, and **SWOT** frameworks to assess Airbnb's external pressures, internal capabilities, competitive position, and strategic vulnerabilities.
- Recommended stronger host/guest verification and AI-driven monitoring to detect fake listings, suspicious activity, off-platform transactions, and quality-control risks.
- Developed a phased action plan across short-term, mid-term, and long-term implementation to improve trust, protect brand reputation, and reduce substitution pressure from hotels and competing platforms.

IT Governance and Consulting — Jefford's Risk Mitigation

Spring 2026

- Led a team of six to design an agile InfoSec management framework and 8-quarter risk mitigation roadmap for Jefford's.
- Mapped initiatives across business continuity planning, disaster recovery planning, laptop encryption, automated malware patching, and fraud protection team establishment.
- Verified deliverables against **COBIT**, **ISO**, and **NIST** standards and presented findings through Gantt charts and phased implementation timelines.
- Addressed regulatory compliance considerations across EU, U.S., and international data jurisdictions.

TARGETED RESUME MAPPING

- **Data Analyst / BI Analyst:** Mehta Papers Internship, Medicaid Analysis, Snowflake Schema, Lending Bias Analysis, Data Center Location Selection, Sentiment Analysis.
- **AI / ML / AI-in-Business:** Marketing Assistant AI Agent, Sentiment Analysis using GCP and PySpark, Heart Disease ML Project, Rumore.
- **Product / Startup / Technical Builder:** Rumore, Marketing Assistant AI Agent, SEMS, Airbnb Strategic Analysis.
- **Cybersecurity / GRC / Risk Analyst:** Jefford's Risk Mitigation, Rumore Authentication Architecture, SEMS Role-Based Access, Airbnb Trust/Safety Analysis.
- **Drone / Defense Tech Adjacent:** Sentiment Analysis using GCP/PySpark, Rumore Backend Architecture, Marketing Assistant AI Agent, Jefford's Risk Mitigation. Add a drone-specific project before using a defense-focused resume.

PORTFOLIO NOTES

- Use this document as a master inventory; do not submit this full version to job applications.
- Convert each project into a portfolio page with: problem, tools, process, screenshots, result, and what I personally contributed.
- For each job application, select only the 3–5 most relevant projects and rewrite the top bullets around the job description.
- Add GitHub, Tableau Public, demo videos, screenshots, or short project writeups wherever available.